A public-message privacy counterexample

STOC submission 1234: the reverse Tetrahedral protocol

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Abstract

The reverse protocol in Theorem 5.1 of Donald Beaver's *Correlated Group Interactions: Founding Cryptography on S_4 and S_n* leaks information about Alice's hand through its public message. Six equiprobable cases suffice to exhibit the failure. This note gives the exact calculation and its correspondence to a checked Lean theorem.

1 Experiment and privacy requirement

The source is Theorem 5.1, PDF page 5, Figures 9 and 10 of the submission manuscript. Alice chooses an ordered pair $A = (a, b)$ of distinct cards from $\{0, 1, 2, 3\}$ uniformly among the twelve possibilities. She publishes M according to the Alice table. An independent uniform OT selection bit determines Bob's card C via Figure 9. For each A, its two OT cases yield the two cards outside A, once each. Thus (a, b, C, D) is uniform over all 24 permutations, where D is the discarded card.

In an ideal $(2, 1; 1)$ deal, Bob receives only C. In the real protocol he also sees M. A necessary privacy condition is that this public component of his view can be simulated from C alone, preserving its joint distribution with Alice's hand. In particular,

$$\Pr[A = a \mid C = c, M = m] = \Pr[A = a \mid C = c]$$

whenever the conditioning event has positive probability. Failure for (C, M) suffices: Bob's other view components need not be examined.

2 Six equiprobable cases

Condition on $C = 0$. The following six Figure 9 cases each have conditional probability $1/6$:

Alice's ordered hand A	Bob C	Discard D	Public M
$(1, 2)$	0	3	Y
$(1, 3)$	0	2	X
$(2, 1)$	0	3	X
$(2, 3)$	0	1	Z
$(3, 1)$	0	2	Z
$(3, 2)$	0	1	Y

Before the message, Bob gives probability $1/6$ to $A = (1, 3)$. Once he sees $M = X$, only $(1, 3)$ and $(2, 1)$ remain, each with probability $1/2$.

3 Exact leakage calculation

We obtain the strict inequality

$$\Pr[A = (1,3) \mid C = 0] = \frac{1}{6} \neq \frac{1}{2} = \Pr[A = (1,3) \mid C = 0, M = X].$$

The defect persists even if the order of Alice's cards is ignored. With $H = \{a, b\}$:

Alice's unordered hand H	$\Pr[H \mid C = 0]$	$\Pr[H \mid C = 0, M = X]$
$\{1,2\}$	$1/3$	$1/2$
$\{1,3\}$	$1/3$	$1/2$
$\{2,3\}$	$1/3$	0

In particular, Bob rules out the previously possible hand $\{2,3\}$. A simulator given only C cannot reproduce this extra correlation with Alice's hand.

4 The checked Lean counterexample

Let E denote $A = (1,3)$, F denote $C = 0$, and G denote $M = X$. Among the 24 shuffles,

$$|E \cap F \cap G| = 1, \qquad |F| = 6, \qquad |E \cap F| = 1, \qquad |F \cap G| = 2.$$

Conditional independence would require

$$|E \cap F \cap G|\,|F| = |E \cap F|\,|F \cap G|.$$

Instead, the two sides are 6 and 2. The declaration

`Cslib.Crypto.Tetrahedral.reverse_message_leakage`

checks precisely this division-free inequality, using `by decide` to enumerate the finite cases. Separate checked declarations establish that the enumerator consists of exactly the 24 distinct permutations. The message encoding is $X = 0$, $Y = 1$, $Z = 2$.

5 Faulty inference and scope

The proof of Theorem 5.1 argues from M being determined by Alice's hand to privacy. Determinism given Alice's hand does not imply independence from that hand given Bob's card. Checking a joint probability only for a message consistent with a fixed hand does not prove the required conditional independence; incompatible hands are exactly what the observed message excludes.

The counterexample contradicts perfect semi-honest privacy for this reverse protocol. It does not refute its dealt-output distribution, the forward protocol's checked correctness/privacy, or the existence of a different secure reverse construction. It is not a claim that the entire paper is false.

Sources and reproducibility

Primary source: manuscript PDF page 5, Figures 9–10 and Theorem 5.1 (linked on page 1). Protocol tables are also Figures 2–3. The immutable Lean proof is in `TetrahedralYield.lean`, lines 106–116, in Complexitylib PR 2529.

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Kernel checking establishes the counting inequality; correspondence to the manuscript additionally uses the explicit table mapping above. AI-assisted technical note; Apache-2.0.